```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetServic
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DS
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DS
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DS
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DS
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber opti

5 rows × 21 columns

```
df.shape
```

```
(7043, 21)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
df = df.drop(['customerID'],axis = 1)
df = df.drop(['customerID'], axis=1, errors='ignore')
df.head()
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSec
0	Female	0	Yes	No	1	No	No phone service	DSL	
1	Male	0	No	No	34	Yes	No	DSL	
2	Male	0	No	No	2	Yes	No	DSL	
3	Male	0	No	No	45	No	No phone service	DSL	
4	Female	0	No	No	2	Yes	No	Fiber optic	

Next steps: Generate code with df | New interactive sheet

```
df.sample(4)
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	Online
6371	Female	0	No	No	24	Yes	Yes	No	N
2241	Female	0	Yes	Yes	56	Yes	Yes	No	N
5609	Male	0	Yes	No	29	Yes	Yes	Fiber optic	
2456	Male	0	Yes	Yes	47	Yes	Yes	No	N

```
features = ['tenure', 'MonthlyCharges', 'TotalCharges']
for col in df.columns:
  if col not in features:
    print(col,df[col].unique())
    print("-"*50)
```

```
gender ['Female' 'Male']
--------------------------------------------------
SeniorCitizen [0 1]
--------------------------------------------------
Partner ['Yes' 'No']
--------------------------------------------------
Dependents ['No' 'Yes']
--------------------------------------------------
PhoneService ['No' 'Yes']
--------------------------------------------------
MultipleLines ['No phone service' 'No' 'Yes']
--------------------------------------------------
InternetService ['DSL' 'Fiber optic' 'No']
--------------------------------------------------
OnlineSecurity ['No' 'Yes' 'No internet service']
--------------------------------------------------
OnlineBackup ['Yes' 'No' 'No internet service']
--------------------------------------------------
DeviceProtection ['No' 'Yes' 'No internet service']
--------------------------------------------------
TechSupport ['No' 'Yes' 'No internet service']
--------------------------------------------------
StreamingTV ['No' 'Yes' 'No internet service']
--------------------------------------------------
StreamingMovies ['No' 'Yes' 'No internet service']
--------------------------------------------------
Contract ['Month-to-month' 'One year' 'Two year']
--------------------------------------------------
PaperlessBilling ['Yes' 'No']
--------------------------------------------------
PaymentMethod ['Electronic check' 'Mailed check' 'Bank transfer (automatic)'
 'Credit card (automatic)']
--------------------------------------------------
Churn ['No' 'Yes']
--------------------------------------------------
```

```
df.isnull().sum()
```

	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

```
df[df['TotalCharges']==" "]
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	Online
488	Female	0	Yes	Yes	0	No	No phone service	DSL	
753	Male	0	No	Yes	0	Yes	No	No	N
936	Female	0	Yes	Yes	0	Yes	No	DSL	
1082	Male	0	Yes	Yes	0	Yes	Yes	No	N
1340	Female	0	Yes	Yes	0	No	No phone service	DSL	
3331	Male	0	Yes	Yes	0	Yes	No	No	N
3826	Male	0	Yes	Yes	0	Yes	Yes	No	N
4380	Female	0	Yes	Yes	0	Yes	No	No	N
5218	Male	0	Yes	Yes	0	Yes	No	No	N
6670	Female	0	Yes	Yes	0	Yes	Yes	DSL	
6754	Male	0	No	Yes	0	Yes	Yes	DSL	

```
df['TotalCharges'] = df['TotalCharges'].replace(" ", '0.0')
df[df['TotalCharges']=="0.0"]
```

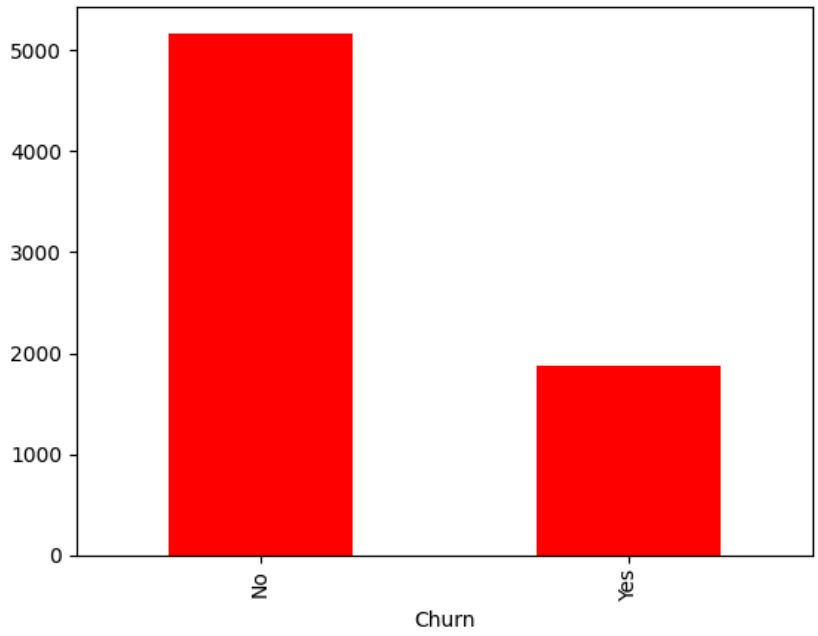
	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	Online
488	Female	0	Yes	Yes	0	No	No phone service	DSL	
753	Male	0	No	Yes	0	Yes	No	No	N
936	Female	0	Yes	Yes	0	Yes	No	DSL	
1082	Male	0	Yes	Yes	0	Yes	Yes	No	N
1340	Female	0	Yes	Yes	0	No	No phone service	DSL	
3331	Male	0	Yes	Yes	0	Yes	No	No	N
3826	Male	0	Yes	Yes	0	Yes	Yes	No	N
4380	Female	0	Yes	Yes	0	Yes	No	No	N
5218	Male	0	Yes	Yes	0	Yes	No	No	N
6670	Female	0	Yes	Yes	0	Yes	Yes	DSL	
6754	Male	0	No	Yes	0	Yes	Yes	DSL	

```
df['TotalCharges'] = df['TotalCharges'].astype(float)
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   gender            7043 non-null   object
 1   SeniorCitizen     7043 non-null   int64
 2   Partner           7043 non-null   object
 3   Dependents        7043 non-null   object
 4   tenure            7043 non-null   int64
 5   PhoneService      7043 non-null   object
 6   MultipleLines     7043 non-null   object
 7   InternetService   7043 non-null   object
 8   OnlineSecurity    7043 non-null   object
 9   OnlineBackup      7043 non-null   object
 10  DeviceProtection  7043 non-null   object
 11  TechSupport       7043 non-null   object
 12  StreamingTV       7043 non-null   object
 13  StreamingMovies   7043 non-null   object
 14  Contract          7043 non-null   object
 15  PaperlessBilling  7043 non-null   object
 16  PaymentMethod     7043 non-null   object
 17  MonthlyCharges    7043 non-null   float64
 18  TotalCharges      7043 non-null   float64
 19  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(16)
memory usage: 1.1+ MB
```

```
df['Churn'].value_counts().plot(kind = 'bar', color='red')
```

```
<Axes: xlabel='Churn'>
```




```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
def kdeplot(feature):
    plt.figure(figsize=(9, 4))
    plt.title(f"KDE for {feature}")

    ax0 = sns.kdeplot(df[df['Churn'] == 'No'][feature].dropna(), color='red', label='Churn: No', shade=True)
    ax1 = sns.kdeplot(df[df['Churn'] == 'Yes'][feature].dropna(), color='blue', label='Churn: Yes', shade=True

    plt.legend()  # legend
    plt.show()  # Ensure the plot is displayed properly

kdeplot('tenure')
kdeplot('MonthlyCharges')
kdeplot('TotalCharges')
```

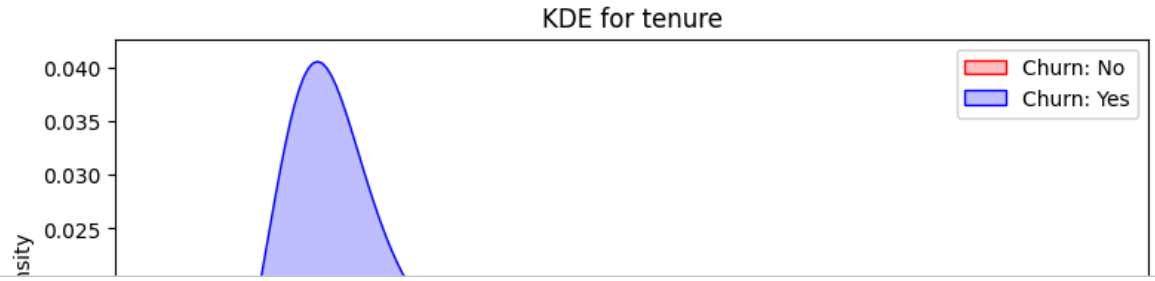
```
/tmp/ipykernel_3429/4202423896.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

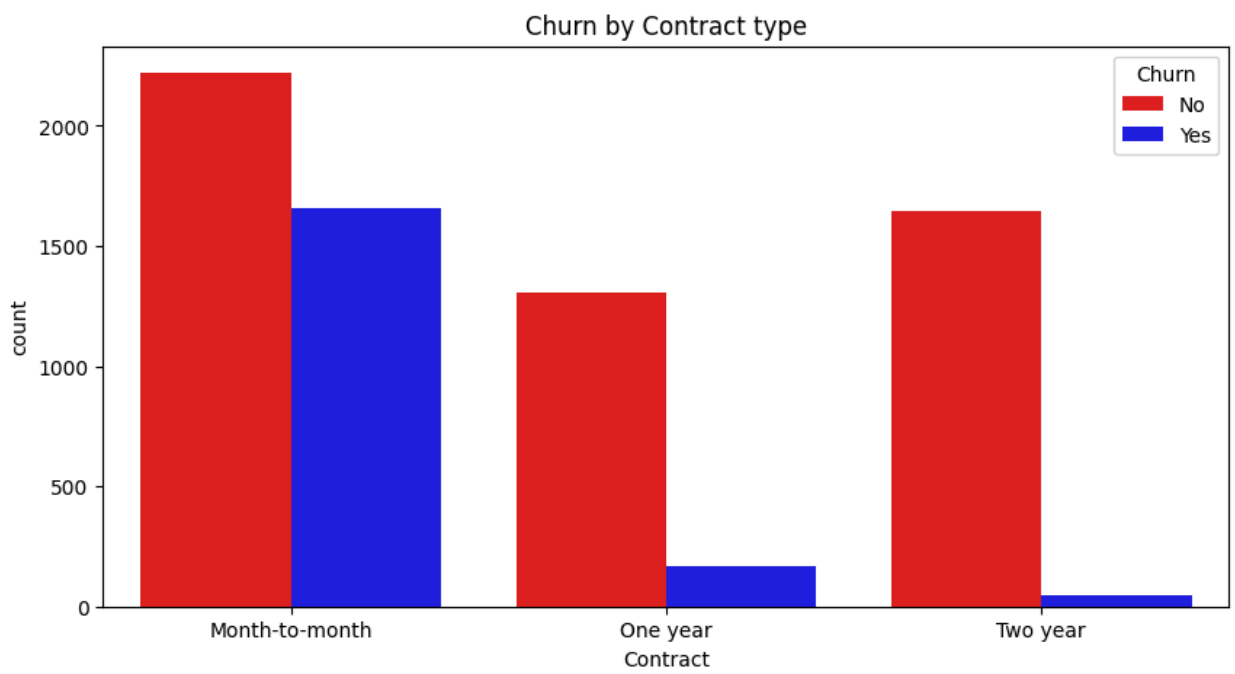
  ax0 = sns.kdeplot(df[df['Churn'] == 'No'][feature].dropna(), color='red', label='Churn: No', shade=True)
/tmp/ipykernel_3429/4202423896.py:6: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

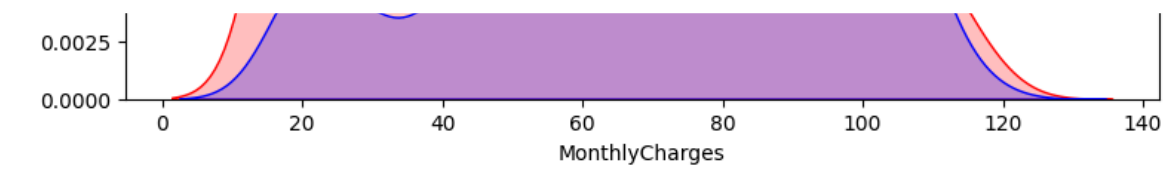
  ax1 = sns.kdeplot(df[df['Churn'] == 'Yes'][feature].dropna(), color='blue', label='Churn: Yes', shade=True)
```




```
plt.figure(figsize=(10,5))
sns.countplot(x='Contract', hue='Churn', data=df, palette=['red', 'blue'])
plt.title('Churn by Contract type')
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.boxplot(x='Contract', y = 'MonthlyCharges', hue='Churn', data=df, palette=['red', 'blue'])
plt.title('Churn by Contract type')
plt.show()
```



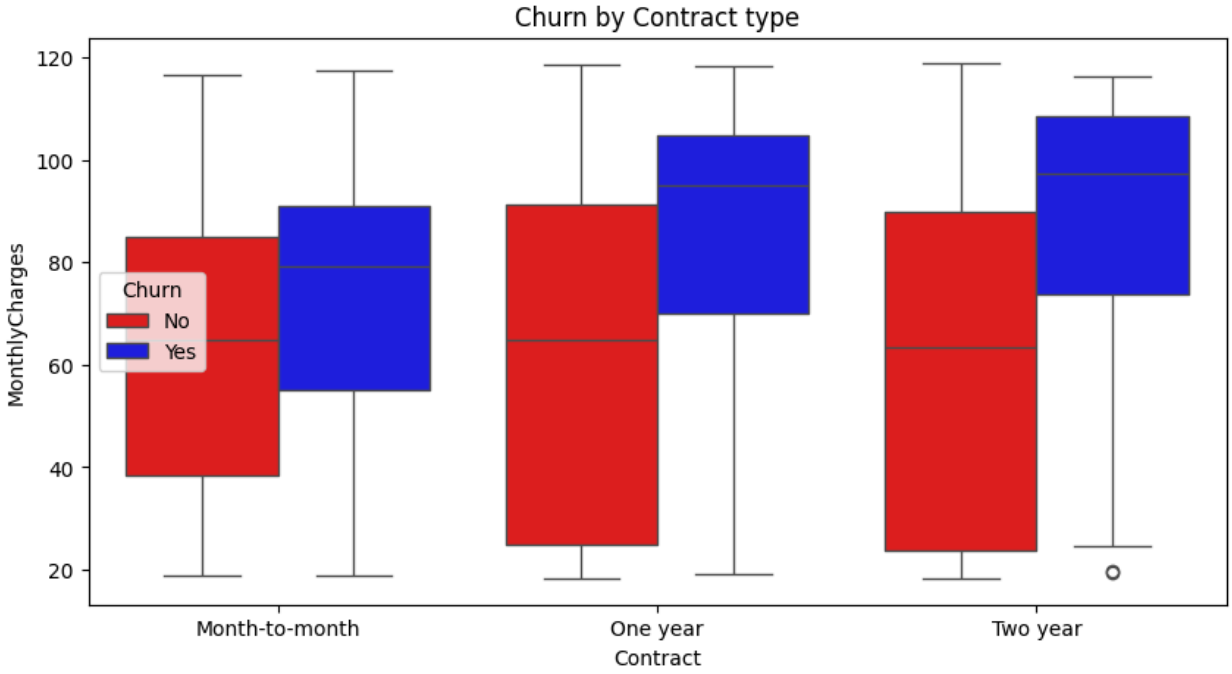

```
/tmp/ipykernel_3429/4202423896.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

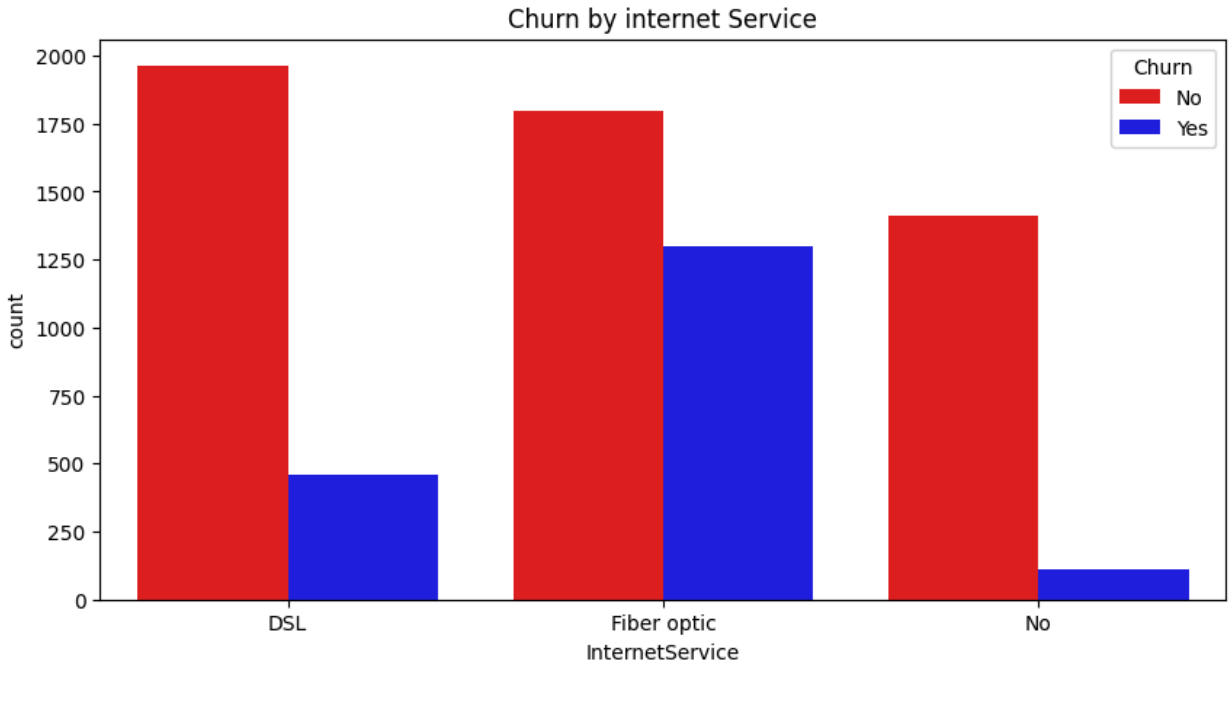
  ax0 = sns.kdeplot(df[df['Churn'] == 'No'][feature].dropna(), color='red', label='Churn: No', shade=True)
/tmp/ipykernel_3429/4202423896.py:6: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

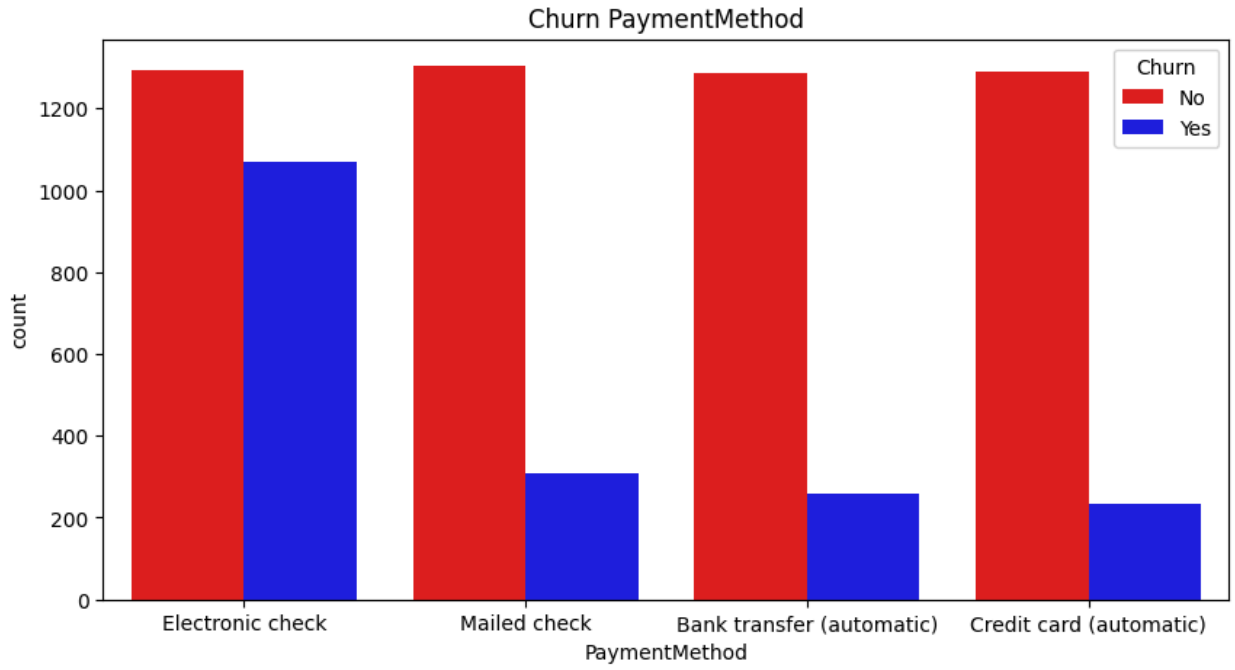
  ax1 = sns.kdeplot(df[df['Churn'] == 'Yes'][feature].dropna(), color='blue', label='Churn: Yes', shade=True)
```



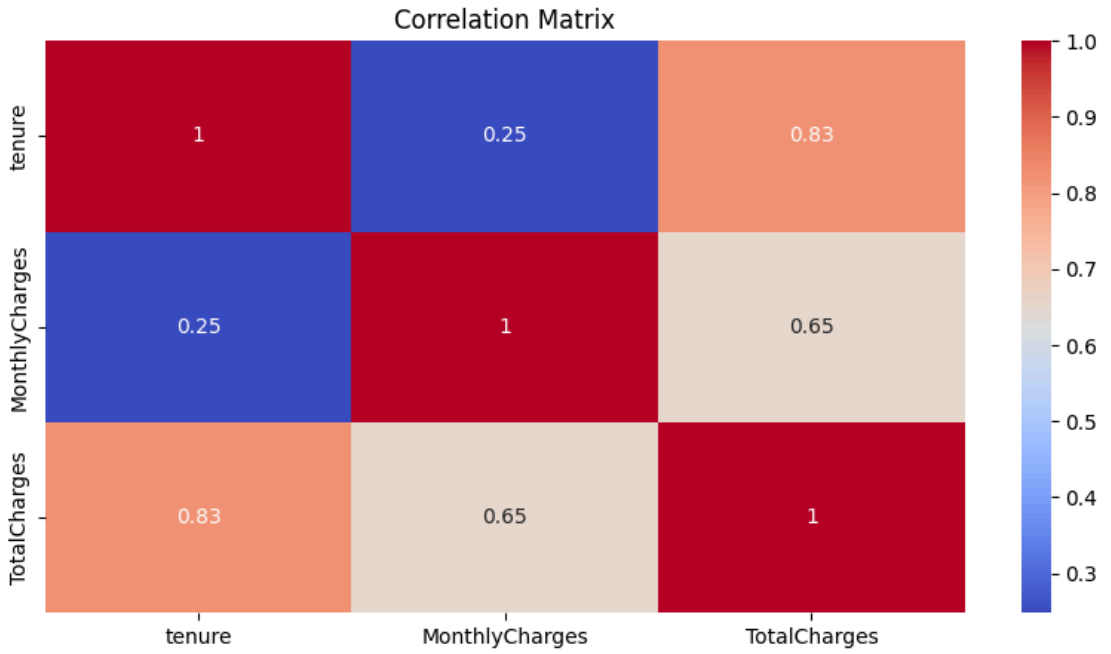

```
plt.figure(figsize=(10,5))
sns.countplot(x='InternetService', hue='Churn', data=df, palette=['red', 'blue'])
plt.title('Churn by internet Service')
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.countplot(x='PaymentMethod', hue='Churn', data=df, palette=['red', 'blue'])
plt.title('Churn PaymentMethod')
plt.show()
```




```
plt.figure(figsize=(10,5))
sns.heatmap((df[['tenure','MonthlyCharges','TotalCharges']]).corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```




```
df['MultipleLines'].replace('No phone service','No',inplace=True)
df['OnlineSecurity'].replace('No internet service','No',inplace=True)
df['OnlineBackup'].replace('No internet service','No',inplace=True)
df['DeviceProtection'].replace('No internet service','No',inplace=True)
df['TechSupport'].replace('No internet service','No',inplace=True)
df['StreamingTV'].replace('No internet service','No',inplace=True)
df['StreamingMovies'].replace('No internet service','No',inplace=True)
```

```
/tmp/ipykernel_3429/3086996000.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True


  df['MultipleLines'].replace('No phone service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or
```

```
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['OnlineSecurity'].replace('No internet service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['OnlineBackup'].replace('No internet service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['DeviceProtection'].replace('No internet service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:5: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['TechSupport'].replace('No internet service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['StreamingTV'].replace('No internet service','No',inplace=True)
/tmp/ipykernel_3429/3086996000.py:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df['StreamingMovies'].replace('No internet service','No',inplace=True)
```

```
yes_no_columns = ['Partner','Dependents','PhoneService','MultipleLines','OnlineSecurity','OnlineBackup',
                  'DeviceProtection','TechSupport','StreamingTV','StreamingMovies','PaperlessBilling','Churn']
for col in yes_no_columns:
    df[col].replace({'Yes': 1,'No': 0},inplace=True)
```

```
/tmp/ipykernel_3429/2444924067.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Se
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object or

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True

  df[col].replace({'Yes': 1,'No': 0},inplace=True)
/tmp/ipykernel_3429/2444924067.py:4: FutureWarning: Downcasting behavior in `replace` is deprecated and will b
  df[col].replace({'Yes': 1,'No': 0},inplace=True)
```

```
df.head()
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSec
0	Female	0	1	0	1	0	0	DSL	
1	Male	0	0	0	34	1	0	DSL	
2	Male	0	0	0	2	1	0	DSL	
3	Male	0	0	0	45	0	0	DSL	
4	Female	0	0	0	2	1	0	Fiber optic	

Next steps: Generate code with df | New interactive sheet

```
df['gender'] = df['gender'].replace({'Female': 1, 'Male': 0})
df.head()
```

```
/tmp/ipykernel_3429/4058162116.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will b
  df['gender'] = df['gender'].replace({'Female': 1, 'Male': 0})
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSec
0	1	0	1	0	1	0	0	DSL	
1	0	0	0	0	34	1	0	DSL	
2	0	0	0	0	2	1	0	DSL	
3	0	0	0	0	45	0	0	DSL	
4	1	0	0	0	2	1	0	Fiber optic	

Next steps: Generate code with df | New interactive sheet

```
df2=pd.get_dummies(data=df, columns=['InternetService', 'Contract', 'PaymentMethod'])
df2.head()
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	OnlineSecurity	OnlineBack
0	1	0	1	0	1	0	0	0	
1	0	0	0	0	34	1	0	1	
2	0	0	0	0	2	1	0	1	
3	0	0	0	0	45	0	0	1	
4	1	0	0	0	2	1	0	0	

5 rows × 27 columns

```
col_to_scale = ['tenure', 'MonthlyCharges', 'TotalCharges']
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df2[col_to_scale] = scaler.fit_transform(df2[col_to_scale])
```

```
df2.dtypes
```

	0
gender	int64
SeniorCitizen	int64
Partner	int64
Dependents	int64
tenure	float64
PhoneService	int64
MultipleLines	int64
OnlineSecurity	int64
OnlineBackup	int64
DeviceProtection	int64
TechSupport	int64
StreamingTV	int64
StreamingMovies	int64
PaperlessBilling	int64
MonthlyCharges	float64
TotalCharges	float64
Churn	int64
InternetService_DSL	bool
InternetService_Fiber optic	bool
InternetService_No	bool
Contract_Month-to-month	bool
Contract_One year	bool
Contract_Two year	bool
PaymentMethod_Bank transfer (automatic)	bool
PaymentMethod_Credit card (automatic)	bool
PaymentMethod_Electronic check	bool
PaymentMethod_Mailed check	bool

dtype: object

```
for col in df2:
    print(f'{col}: {df2[col].unique()}')
```

```
gender: [1 0]
SeniorCitizen: [0 1]
Partner: [1 0]
Dependents: [0 1]
tenure: [0.01388889 0.47222222 0.02777778 0.625      0.11111111 0.30555556
 0.13888889 0.38888889 0.86111111 0.18055556 0.22222222 0.80555556
 0.68055556 0.34722222 0.95833333 0.72222222 0.98611111 0.29166667
 0.16666667 0.41666667 0.65277778 1.         0.23611111 0.375
 0.06944444 0.63888889 0.15277778 0.97222222 0.875      0.59722222
 0.20833333 0.83333333 0.25       0.91666667 0.125      0.04166667
 0.43055556 0.69444444 0.88888889 0.77777778 0.09722222 0.58333333
 0.48611111 0.66666667 0.40277778 0.90277778 0.52777778 0.94444444
 0.44444444 0.76388889 0.51388889 0.5        0.56944444 0.08333333
 0.05555556 0.45833333 0.93055556 0.31944444 0.79166667 0.84722222
 0.19444444 0.27777778 0.73611111 0.55555556 0.81944444 0.33333333
 0.61111111 0.26388889 0.75       0.70833333 0.36111111 0.
 0.54166667]
PhoneService: [0 1]
MultipleLines: [0 1]
OnlineSecurity: [0 1]
OnlineBackup: [1 0]
DeviceProtection: [0 1]
TechSupport: [0 1]
```